CLICK HERE CLICK HERE

APUL

NEET FLIX

AIM FREE EDUCATION TO ALL

NEET CHANNEL

JOIN OUR TELEGRAM CHANNELS HERE

TEST PAPERS <<< CLICK

TOPPER'S NOTES <<< CLICK

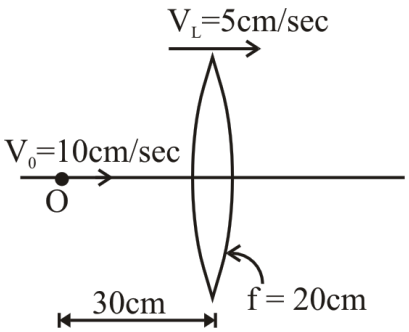
BEST FROM KOTA LECTURES <<< CLICK

QUIZES <<< CLICK

REVISION FLASH CARDS &
MANY MORE <<< CLICK

PHYSICS

1) find velocity of image :



(1) 25cm/sec

(2) 10cm/sec

(3) 40cm/sec

(4) None of the above

2) A beam of light converges to a point P. A lens is placed in path of convergent beam 12 cm from P. At what point does beam converges if lens is concave lens of focal length 16 cm :-

(1) 16 cm

(2) 48 cm

(3) 7.5 cm

(4) 15 cm

3) A concave and a convex lens have the same focal length of 20 cm and are put into contact to form a lens combination. The combination is used to view an object of 5 cm length kept at 20 cm from the lens combination. As compared to the object, the image will be :-

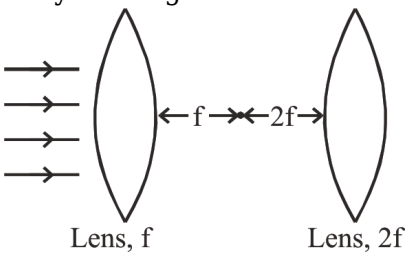
(1) Magnified and inverted

(2) Reduced and erected

(3) Of the same size as the object and erected

(4) Of the same size as the object but inverted

4) Two convex lens of focal length f and 2f are arranged to have their foci at the same point as shown in figure. A circular beam of parallel light rays having 1 cm cross section radius is incident on first lens. The beam exiting the second lens is:-



(1) Diverging

(2) Converging

(3) Parallel and 1 cm cross section radius

(4) Parallel and of 2 cm cross section radius

5) A point source of light is present at the surface of a sphere. A parallel beam of light emerge out from the opposite surface of the sphere. The refractive index of the material of the sphere is

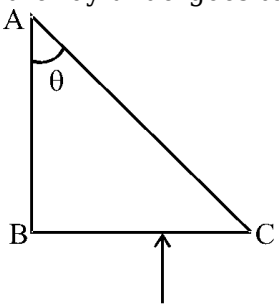
(1) 1.5

(2) 5/3

(3) 2

(4) 2.5

6) A ray of light is incident on a right angle glass prism (μ = 3/2) at right angles to face BC. Calculate the maximum value for angle θ so that the ray undergoes total internal reflection at face



AC when the prism is placed in water (μ = 4/3):-

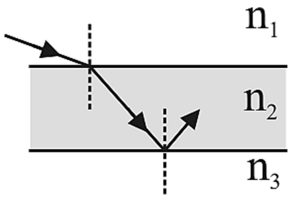
(1) 35°

(2) 45°

(3) 30°

(4) 27°

7) Light traveling through three transparent substances follows the path shown in figure. Arrange the indices of refraction in order from smallest to largest. Note that total internal reflection does



occur on the bottom surface of medium 2.

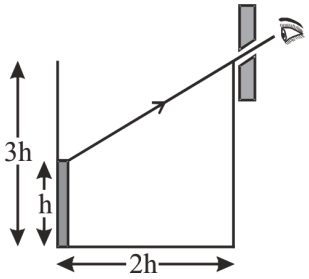
(1) $n_1 < n_2 < n_3$

(2) $n_2 < n_1 < n_3$

(3) $n_1 < n_3 < n_2$

(4) $n_3 < n_1 < n_2$

8) An observer can see through a pin-hole the top end of a thin rod of height h, placed as shown in the figure. The beaker height is 3h and its radius h. When the beaker is filled with a liquid up to a height 2h, he can see the lower end of the rod. Then the refractive index of the liquid is :-



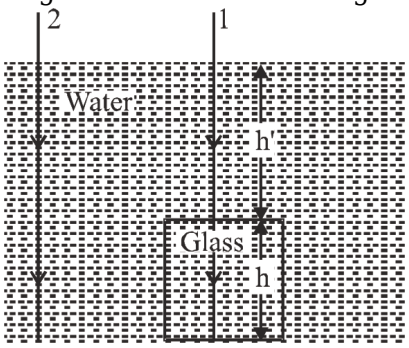
(1) 5/2

(2) $\sqrt{(5/2)}$

(3) $\sqrt{(3/2)}$

(4) 3/2

9) Two beams of light are incident normally on water (R.I. = 4/3). If the beam 1 passes through a glass (R.I. = 3/2) slab of height h as shown in the figure, the time difference for both the beams for reaching the bottom is :-



(1) zero

(2) $\frac{h'}{6c}$

(3) $\frac{6h}{c}$

(4) $\frac{h}{6c}$

10) A ray of light is incident at an angle of incidence 60° on the glass slab of refractive index $\sqrt{3}$. After refraction, the light ray emerges out from other parallel faces and lateral shift between incident ray and emergent ray is $4\sqrt{3}$ cm. The thickness of the glass slab is :-

(1) 4 cm

(2) 6 cm

(3) 8 cm

(4) 12 cm

11) The distance of an object from the focus of a convex mirror of radius of curvature 'a' is 'b'. Then the distance of the image from the focus is :

(1) $b^2 / 4a$

(2) a / b^2

(3) $a^2 / 4b$

(4) $4b / a^2$

12) A rod 1.5 m long is placed along the principal axis of a convex mirror of focal length 60cm. The near end of the rod is 1m from the pole of the convex mirror. The length of the image of the rod is

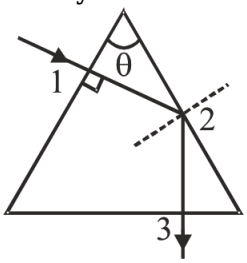
(1) 15 cm

(2) 14.44 cm

(3) 6.62 cm

(4) 10.88 cm

13) A ray of light enters into the prism normally with the face 1 and comes out from the face 3. The minimum refractive index of the prism is



(1) $\operatorname{cosec} \frac{\theta}{2}$

(2) tan θ

(3) sec θ

(4) cosec θ

14) **Assertion :** It is possible to obtain real enlarged image from a convex mirror. **Reason :** Convex mirror may converge or diverge the light incident on it.

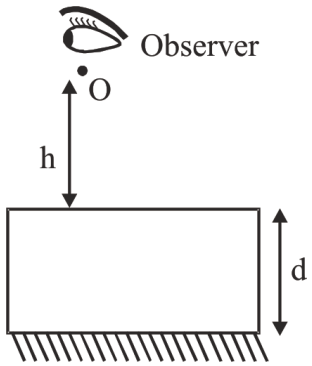
(1) Both Assertion & Reason are True & the Reason is a correct explanation of the Assertion.

(2) Both Assertion & Reason are True but Reason is not a correct explanation of the Assertion.

(3) Assertion is True but the Reason is False.

(4) Both Assertion & Reason are False.

15) A point luminous object (O) is at a distance h from front face of a glass slab of width d and of refractive index μ. Back face of the slab is a reflecting plane mirror. An observer sees the image of object in mirror as shown in the figure. Distance of image from front face as seen by the observer, will be :-



(1) $h + \frac{2d}{\mu}$

(2) 2h +2d

(3) h + d

(4) $h + \frac{d}{\mu}$

16) Read the following statements :
(a) Both concave and convex mirror can form **Virtual** image for Real object.
(b) A **Virtual** image cannot be achieved on a screen but it can be photographed.
(c) The **diver** under water looks obliquely, at a fisherman standing on the bank of a lake, the fisherman look taller to the diver, than what he actually is
(d) The refractive index of diamond is much greater than that of **original** glass
select the correct statements :

(1) a, b

(2) b, c

(3) a, b, c, d

(4) a, b, d

17) Two slits separated by a distance of 1 mm, are illuminated with red light of wavelength 6.5×10^{-7} m. The interference fringes are observed on a screen placed 1 m from the slits. Find the distance between the third dark fringe and the fifth bright fringe on the same side of the central maxima.

(1) 1.625 mm

(2) 3.50 mm

(3) 2.750 mm

(4) None of the above

18) At what angle of incidence, the light reflected from a glass slab will become completely polarised, if the angle of refraction at that incident angle is 33.6°?

(1) 90°

(2) 0°

(3) 56.4°

(4) 46.4°

19) In YDSE, the intensity of light at a point where path difference is λ is 0.01 W/m^2. Find intensity at a point where path difference is λ/6.

(1) 0.0075 W/m^2

(2) 0.0050 W/m^2

(3) 0.0025 W/m^2

(4) None of these

20) A mixture of light, consisting of wavelength 590 nm and an unknown wavelength, illuminates Young's double slit and gives rise to two overlapping interference patterns on the screen. The central maximum of both lights coincide. Further, it is observed that the third bright fringe of known

light coincides with the 4th bright fringe of the unknown light. From this data, the wavelength of the unknown light is :-

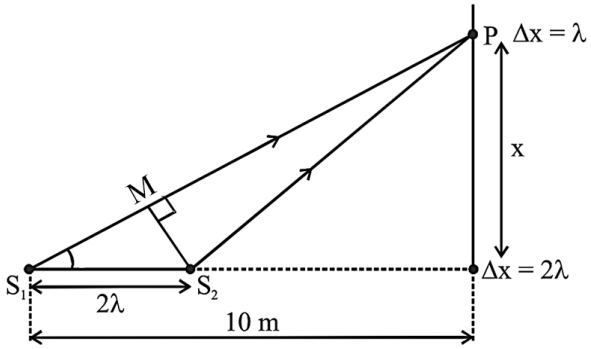
(1) 442.5 nm

(2) 776.8 nm

(3) 393.4 nm

(4) 885.0 nm

21) Find the value of x for which there is maximum Intensity at point P.



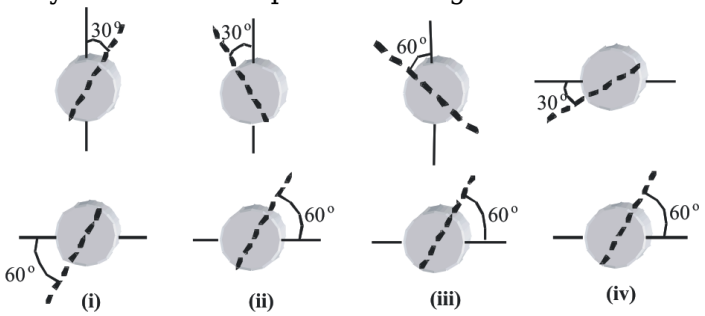
(1) 10 m

(2) $10\sqrt{3}$m

(3) $5\sqrt{3}$m

(4) 5m

22) The figure shows four pairs of polarizing sheets, seen face-on. Each pair is mounted in the path of initially unpolarized light. The polarizing direction of each sheet (indicated by the dashed line) is referenced to either a horizontal x-axis or a vertical y axis. Rank the pair according to the fraction of the initial intensity that they pass, greatest first



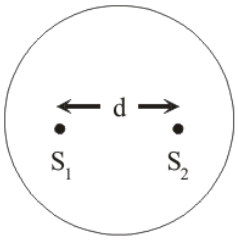
(1) (i) > (ii) > (iii) > (iv)

(2) (i) > (iv) > (ii) > (iii)

(3) (i) > (iii) > (ii) > (iv)

(4) (iv) > (iii) > (ii) > (i)

23) Two coherent sources separated by distance d are radiating in phase having wavelength λ. A detector moves in a big circle around the two sources in the plane of the two sources. The angular



position with line joining two sources of n = 4 interference maxima is given as

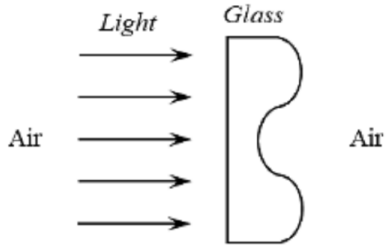
(1) $\sin^{-1}\frac{n\lambda}{d}$

(2) $\cos^{-1}\frac{4\lambda}{d}$

(3) $\tan^{-1}\frac{d}{4\lambda}$

(4) $\cos^{-1}\frac{\lambda}{4d}$

24) A parallel beam of light strikes a piece of transparent glass having cross section as shown in the figure below. Correct shape of the emergent wavefront will be (figures are schematic and not drawn to scale)-



(1)



(2)



(3)



(4) 

25) Which one of the following statements is NOT true for de-broglie waves ?

(1) All atomic particles in motion have waves of a definite wavelength associated with them
(2) The higher the momentum, the longer is the wavelength
(3) The faster the particle, the shorter is the wavelength
(4) For the same velocity, a heavier particle has a shorter wavelength

26) $^{239}_{94}Pu \rightarrow {}^{235}_{92}U + {}^{4}_{2}He$
Total disintegrated energy in this reaction is denoted by Q. Percentage of Q carried away by α particle $(^{4}_{2}He)$ is :-

(1) 9.8%
(2) 98.32%
(3) 1.8%
(4) 90.12%

27) Assuming that 200 MeV of energy is released per fission of $_{92}U^{235}$ atom. Find the number of fission per second required to release 2 kW power :-

(1) 3.125×10^{13}
(2) 6.25×10^{13}
(3) 3.125×10^{15}
(4) 6.25×10^{14}

28) The binding energy per nucleon of Cu^{63} is 8.75 MeV and of Cu^{64} is 8.73 MeV. The energy (in MeV) required is remove a neutron from Cu^{64} is :-

(1) 12.6
(2) 3.64
(3) 4.23
(4) 7.47

29) The minimum frequency of a γ-ray that causes a deuteron to disintegrate into a proton and a neutron is (m_d = 2.0141 amu, m_p = 1.0078 amu, m_n = 1.0087 amu):-

(1) 2.7×10^{20} Hz
(2) 5.4×10^{20} Hz
(3) 10.8×10^{20} Hz

(4) 21.6×10^{20} Hz

30) In an experiment of photo electric effect, when light of certain frequency and intensity is falling on a photocell, 36 mA of photo-current is observed. If the frequency and intensity of incident radiation is made 3 times and 4 times respectively then new photo-current would be, or same each photon emits an electron :-

(1) 144 mA

(2) 96 mA

(3) 48 mA

(4) 27 mA

31) There are two sources of light, each emitting with a power of 100 W. One emits X-rays of wavelength 1 nm and the other visible light of 500 nm. The ratio of number of photons of X-rays to the photons of visible light of the given wavelength is :

(1) 1 : 500

(2) 1 : 400

(3) 1 : 300

(4) 1 : 200

32)

Threshold wavelength of a metal surface is 5×10^{-7} m. When it is illuminated by light of wavelength 3×10^{-7}m , stopping potential is V_0. What will be stopping potential, if wavelength of incident light is doubled :-

(1) $\frac{V_0}{2}$

(2) $2V_0$

(3) V_0

(4) Zero

33) A metal surface is illuminated by light of two different wavelengths 248 nm and 310 nm. The maximum speeds of the photoelectrons corresponding to these wavelengths are u_1 and u_2 respectively. If the ratio $u_1 : u_2 = 2 : 1$ and $hc = 1240$ eV nm, the work function of the metal is nearly :-

(1) 3.7 eV

(2) 3.2 eV

(3) 2.8 eV

(4) 2.5 eV

34)

Light described at a place by the equation
$E=(100\ V/m)[\sin(5\times10^{15}s^{-1})t + \sin(8\times10^{15}s^{-1})t]$ falls on a metal surface having work function 2eV. Calculate the maximum kinetic energy of the photo electrons.

(1) 5.27 eV

(2) 1.71 eV

(3) 3.27 eV

(4) 0.854 eV

35) How much mass is lost by a nuclear reactor operated at 500 MW power level in 1 hour. (Assume 50% efficiency for conversion of nuclear energy into electricity).

(1) 0.04 gm/hr

(2) 0.02 gm/hr

(3) 0.06 gm/hr

(4) 0.07 gm/hr

36) Consider the nuclear fission
$Ne^{20} \rightarrow 2He^{4} + C^{12}$
Given that the binding energy/nucleon of
Ne^{20}, He^{4} and C^{12} are, respectively, 8.03 MeV, 7.07 MeV and 7.68 MeV, identify the correct statement :

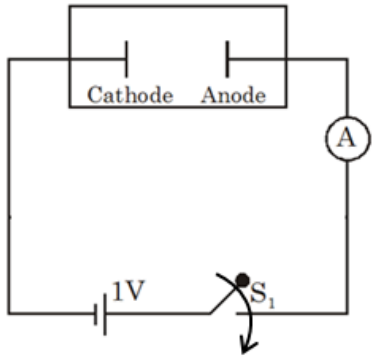
(1) 8.3 MeV energy will be released

(2) energy of 12.4 MeV will be supplied

(3) energy of 11.9 MeV has to be supplied

(4) energy of 3.6 MeV will be released

37) **Answer the following by appropriately matching the lists based on the information given in the paragraph.**
A photo electric cell is shown in the figure. The emitter plate (cathode) is made of copper (ϕ = 4.7 eV) and emitter is illuminated by a source of monochromatic light of wavelength 200 nm. Saturation current is observed when potential difference between cathode and anode is 1V and anode (collector) at higher potential. [Range of ammeter is large] [Take hc = 1240 (ev)nm] (S_1 is closed)



List-I		List-II	
(I)	Minimum kinetic energy of photoelectrons when reaching the collector is	(P)	2.5 eV
(II)	Maximum kinetic energy of photoelectrons when reaching the collector is	(Q)	Remain same

(III)	If wavelength of incident light is decreased (no. of incident photon same), then reading of current as shown by ammeter will	(R)	1 eV
(IV)	If number of incident photons increases, then reading of current as shown by ammeter will	(S)	Increase
		(T)	0
		(U)	Decrease

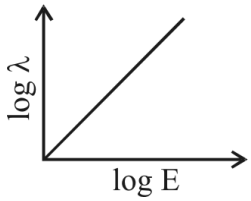
(1) I→P; II→Q; III→S; IV→Q

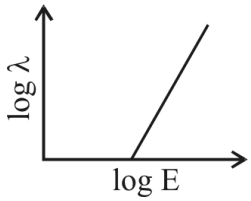
(2) I→R; II→P; III→Q; IV→S

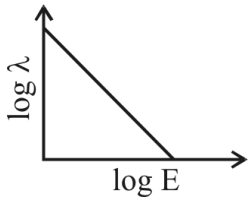
(3) I→S; II→T; III→P; IV→U

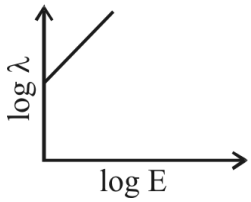
(4) I→T; II→T; III→S; IV→Q

38) The log E - logλ graph between the energy E of an electron and its de-Broglie wavelength λ will be :-

(1) 

(2) 

(3) 

(4) 

39) ***Assertion (A) :-*** Thermonuclear fusion reactions may become the source of unlimited power for the mankind. ***Reason (R) :-*** A single fusion event involving isotopes of hydrogen produces more energy then energy from nuclear fission of single $^{235}_{93}U$.

(1) Both **(A)** and **(R)** are correct but **(R)** is not the correct explanation of **(A)**

(2) **(A)** is correct but **(R)** is not correct

(3) **(A)** is incorrect but **(R)** is correct

(4) Both **(A)** and **(R)** are correct and **(R)** is the correct explanation of **(A)**

40) The maximum wavelength of incident radiation which can create electron-hole pair in pure Ge crystal is :-

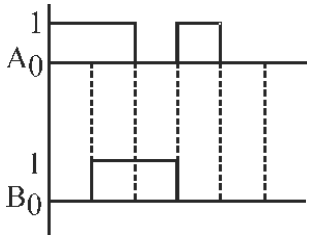
(1) 17714.3 nm

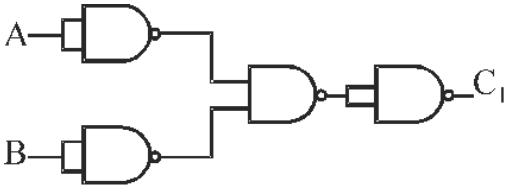
(2) 1771.43 nm

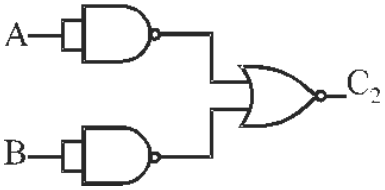
(3) 177.143 nm

(4) 177143 nm

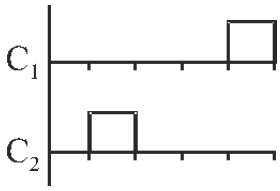
41) Draw the output signal C_1 and C_2 in the given combination of gates :-



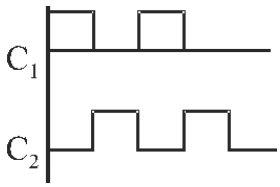




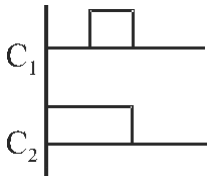
(1)



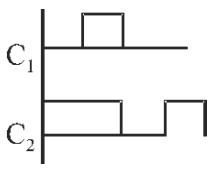
(2)



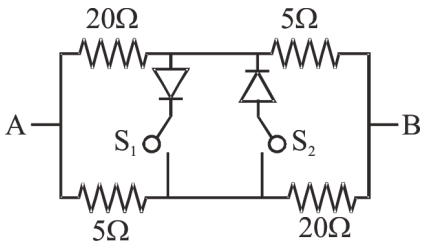
(3)



(4)



42) In the circuit shown below V_A and V_B the potentials at A and B, R is equivalent resistance



between A and B, S_1 and S_2 are switches, and the diodes are ideal :-

(1) If $V_A > V_B$, S_1 is open and S_2 is closed then R = 8Ω

(2) If $V_A > V_B$, S_1 is closed and S_2 is open that R = 12.5Ω

(3) If $V_A < V_B$, S_1 is open and S_2 is closed then R = 12.5Ω

(4) All the above

43) A sinusoidal voltage of peak to peak value of 310 V is connected in series with a diode and a load resistance R so that Half-wave rectification occurs. If the diode has a negligble forwed resistance, the root mean square voltage across the load resistance is

(1) 310 V

(2) 155 V

(3) 109.5 V

(4) 77.5

44) During regulation action of a Zener diode correct statement is :-
(a) the current and voltage across the Zener remains fixed.
(b) the current through the series resistance (R_S) changes.
(c) the Zener resistance is constant.
(d) the resistance offered by the Zener diode changes.

(1) a and c

(2) a and d

(3) b and c

(4) b and d

45) **Assertion :** NAND or NOR gates are called digital building blocks.
Reason : The repeated use of NAND (or NOR) gates can produce all the basic or complicated gates.

(1) If both assertion and reason are true and the reason is the correct explanation of the assertion.

(2) If both assertion and reason are true but reason is not the correct explanation of the assertion.

(3) If assertion is true but reason is false.

(4) If the assertion and reason both are false.

ANSWER KEYS

PHYSICS

Q.	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18	19	20
A.	**1**	**2**	**3**	**4**	**3**	**4**	**4**	**2**	**4**	**4**	**3**	**4**	**4**	**2**	**1**	**3**	**1**	**3**	**1**	**1**
Q.	21	22	23	24	25	26	27	28	29	30	31	32	33	34	35	36	37	38	39	40
A.	**2**	**2**	**2**	**1**	**2**	**2**	**2**	**4**	**2**	**3**	**1**	**4**	**1**	**3**	**1**	**3**	**2**	**3**	**2**	**2**
Q.	41	42	43	44	45															
A.	**1**	**4**	**4**	**4**	**1**															

SOLUTIONS

PHYSICS

1)

Explain Question : We have to find velocity of image for the given condition.

Concept : This question is based on velocity of image in lens.

Solution : As we know that

$V_{\square} = m^2 V_{\square}$

$$v_I - v_L = \frac{v^2}{u^2}(v_o - v_L)$$

$v_I - v_L = m^2 (v_o - v_L)$

$$v_I - v_L = \left(\frac{f}{f+u}\right)^2 (v_o - v_L)$$

$$v_I - 5 = \left(\frac{20}{20-30}\right)^2 (10-5)$$

$$v_I = \left(\frac{20}{-10}\right)^2 (5) + (5)$$

$v_I = 4 \times 5 + 5$

$v_I = 25$ m/s

Final Answer : option (1)

2)

Explain the Question : The question asks where a converging beam of light will meet after passing through a concave lens placed 12 cm before its original focus point P.

Concept : This question is based on lens formula.

Solution : Given u = +12 cm (virtual object, since rays converge to P),
f = –16 cm(concave lens). Using the lens formula :

$$\frac{1}{v} - \frac{1}{u} = \frac{1}{f}$$

$$\frac{1}{v} - \frac{1}{+12} = \frac{1}{-16}$$

$$\frac{1}{v} = -\frac{1}{16} + \frac{1}{12}$$

v = 48 cm

Final answer : 2

3)

Equivalent focal length

$$\frac{1}{F} = \frac{1}{20} - \frac{1}{20} \Rightarrow F = \infty$$

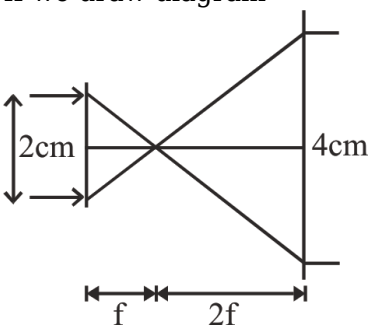
act as glass slab

So image formed is of same size and erected.

4) **Explain Question:**
A parallel beam of radius 1 cm enters a lens of focal length f. A second lens of focal length $2f$ shares the same focal point. We have to find the nature of the emergent beam and its radius.

Concept:
Parallel rays → meet at focal point
Rays from focal point → emerge parallel from the lens.

Solution:
If we draw diagram



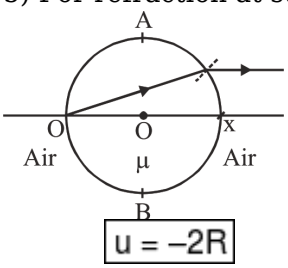
from the diagram, the incident parallel beam has radius:
Radius = 1cm ⇒ Diameter = 2cm
This forms a similar triangle with the ray extending from the common focal point to the second lens.
Distances shown:
* From first lens to focus = f
* From focus to second lens = $2f$
Using similar triangles:

$$\frac{\text{Final diameter}}{\text{Initial diameter}} = \frac{2f}{f} = 2$$

so final diameter = 2 × 2cm = 4cm
⇒ Final radius = 2cm
Since rays reach the second lens form its focal point, the emergent beam is parallel.

Final answer: Option (4)

5) For refraction at surface AB,



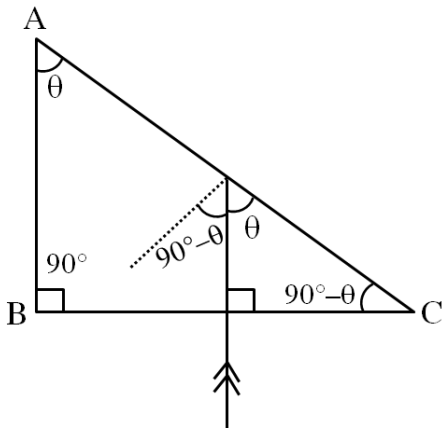
$u = -2R$

$v = +\infty$ $(m_1 = m \Rightarrow m_2 = 1)$
So, applying

$$\Rightarrow \frac{n_2}{v} - \frac{n_1}{v} = \frac{n_2 - n_1}{R}$$

$$\Rightarrow \mu = 2\mu - 2$$

$\therefore \quad \mu = 2$



6)

$90 - \theta > i_c$

$\sin(90 - \theta) > \sin i_c$

$\cos\theta > \frac{4/3}{3/2}$

$\cos\theta > \frac{8}{9}$

7)

Key Concepts:

Snell's Law: When light travels from one medium to another, the relationship between the angle of incidence (θ_1) and the angle of refraction (θ_2) is given by Snell's Law:

$n_1 \sin(\theta_1) = n_2 \sin(\theta_2)$

• where n_1 and n_2 are the refractive indices of the two media.

• **Total Internal Reflection:** Total internal reflection occurs when light travels from a denser medium (higher refractive index) to a rarer medium (lower refractive index) and the angle of incidence exceeds the critical angle.

• **Medium 1 to Medium 2:** The light ray bends away from the normal as it enters medium 2. This indicates that the refractive index of medium 2 (n_2) is less than that of medium 1 (n_1).

8) **Explain Question:**

As shown in figure we have to find refractive index of the liquid.

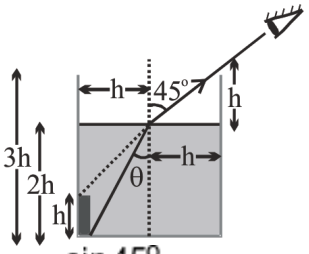
Concept:

Normal shift.

Solution:

The line of sight of the observer remains constant, making angle of 45° with the normal.

$$\sin\theta = \frac{h}{\sqrt{h^2 + (2h)^2}} = \frac{1}{\sqrt{5}}$$



$$\mu = \frac{\sin 45^o}{\sin \theta}$$

$$= \frac{1/\sqrt{2}}{1/\sqrt{5}} = \sqrt{\left(\frac{5}{2}\right)}$$

Final Answer: Option (2)

9)

$v = \frac{c}{\mu}$, $h = \left(\frac{c}{3/2}\right) t_1$ ....(1)

$h = \frac{c}{(4/3)} t_2 \Rightarrow t_1 - t_2 = \left(\frac{3}{2} - \frac{4}{3}\right)\frac{h}{c}$

$t_1 - t_2 = \frac{h}{6c}$

10) **Explain the Question :** To find the thickness of glass slab if incidence angle is 60°, $\mu = \sqrt{3}$, lateral shift is $4\sqrt{3}$ cm.

Concept : This question is based on lateral shift by slab.

Solution :

$\mu_1 \sin i = \mu_2 \sin r$

$1 \sin 60° = \sqrt{3} \sin r$

$\sin r = \frac{1}{2}$

$r = 30°$

Lateral shift $x = \frac{t \sin(i-r)}{\cos r}$

$4\sqrt{3} = \frac{t \sin(60^o - 30^o)}{\cos 30^o} = t \tan 30^o$

$4\sqrt{3} = \frac{t}{\sqrt{3}}$

$t = 12$ cm

Final Answer : 4

11) **Explain the Question :** To find the distance of image from the focus if the distance of on object from focus in at 'b' distance.

Concept : This question is based on newton's formula.

Solution : $F^2 = x_1 \times x_2$

$F = \frac{R}{2} = \frac{a}{2}$

Putting values

$$\left(\frac{a}{2}\right)^2 = b \times x_2$$

$$x_2 = \frac{a^2}{4b}$$

Final Answer : 3

12)

Explain Question : Find length of Image.

Concept : This question is based on longitudinal magnification.

Solution :

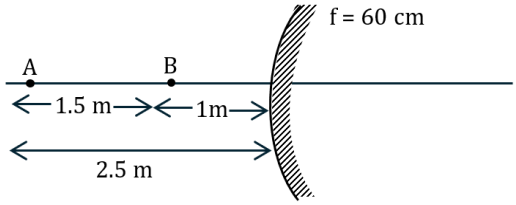


Image of point B

$$\frac{1}{v} + \frac{1}{u} = \frac{1}{f}$$

$$\frac{1}{v} = \frac{1}{f} - \frac{1}{u}$$

$$\frac{1}{v_B} = \frac{1}{60} - \frac{1}{-100}$$

$$\frac{1}{v_B} = \frac{1}{60} + \frac{1}{100}$$

$$V_B = \frac{6000}{160} = 37.5$$

Image of point A

$$\frac{1}{v_A} = \frac{1}{60} - \frac{1}{-250}$$

$$\frac{1}{v_A} = \frac{1}{60} + \frac{1}{250}$$

$$V_A = \frac{15000}{310} = 48.38$$

Length of image =

$$\Rightarrow (V_A - V_B)$$

$\Rightarrow$ 48.38 – 37.5

$\Rightarrow$ 10.88 cm

Final Answer : option (4)

13) Snell's law at 2, for TIR of the ray:

$\sin\theta \leqslant \frac{1}{n}$ or n ≥ cosec θ

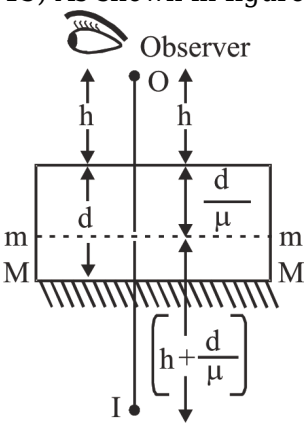
14)

Generated by Allie

Problem Statement: The question asks whether a convex mirror can produce a real,

enlarged image and whether the reason that convex mirrors can both converge or diverge light is true or explains the assertion correctly.
Underlying Concept: A convex mirror always causes light rays to diverge and forms images that are virtual, erect, and smaller than the object. Real images are formed when light rays actually converge. Since convex mirrors cannot converge light, they do not form real images, especially not enlarged real images.
Tips and Tricks: Remember that convex mirrors always diverge rays and produce virtual, diminished images irrespective of object position.
Common Mistakes: Assuming convex mirrors can form real images or that they can converge light.
Why Other Options Are Incorrect?: Choices that state the assertion is true are incorrect because convex mirrors cannot produce real images. Also, reasons suggesting convergence of light by convex mirrors are wrong because they only diverge light.

15) As shown in figure



glass slab will form the image of bottom i.e., mirror MM' at a depth $\left(\frac{d}{\mu}\right)$ from its front face. So the distance of object O from virtual mirror mm' will be h + $\left(\frac{d}{\mu}\right)$.
Now as a plane mirror forms image behind the mirror at the same distance as the object is in front of it, the distance of image I from mm' will be h + $\left(\frac{d}{\mu}\right)$ and as the distance of virtual mirror from the front face of slab is $\left(\frac{d}{\mu}\right)$, the distance of image I from front face as seen by observer will be

$$= \left[h + \frac{d}{\mu}\right] + \frac{d}{\mu} = h + \frac{2d}{\mu}$$

16)

Explain the Question : We need to identify the correct statement / statements from the given.
Concept : This question is based on reflection and refraction of light and image formation by mirror and its properties.
Solution :
(a) True — concave mirror forms virtual image for real object when object is between focus and pole; convex always forms virtual image.
(b) True — virtual image cannot be achieved on screen but can be photographed.

(c) True — the diver sees the fisherman taller, because of refraction at water surface.
(d) True — refractive index of diamond (~2.42) is greater than that of ordinary glass (~1.5).
Hence correct statements are (a), (b), c, (d).
Final answer : 3

17)

Position of dark fringe $\left((2n-1)\frac{\lambda D}{2d}\right)$

Position of bright fringe $\left(\frac{n\lambda D}{d}\right)$

Difference between 3rd dark and 5th bright fringe $= \frac{5\lambda D}{d} - \frac{5\lambda D}{2d}$

$= \frac{\lambda D}{d}\left(5 - \frac{5}{2}\right)$

$= \frac{5}{2} \times \frac{6.5 \times 10^{-7} \times 1}{10^{-3}} = 1.625\text{mm}$

18)

To find at what angle of incidence will light reflected from a glass slab become completely polarised, if angle of refraction is 33.6°
At Brewster angle :
i + r = 90°
i = 90 - 33.6°
= 56.4°

19) $\Delta x = \frac{\lambda}{6} \quad \phi = \frac{2\pi}{6} = \frac{\pi}{3}$ or 60°

$I_1 = I_2 = I'$

$I_{max} = 4I' = 0.01$

$I' = \frac{0.01}{4}$

$I = I_1 + I_2 + 2\sqrt{I_1}\sqrt{I_2}\cos\phi$

$= \frac{0.01}{4} + \frac{0.01}{4} + 2\sqrt{\frac{0.01}{4}}\sqrt{\frac{0.01}{4}}\cos 60^\circ$

$= \frac{0.01}{4} + \frac{0.01}{4} + 2 \times \frac{0.01}{4} \times \frac{1}{2} = \frac{0.03}{4}$ W/m^2 = 0.0075 W/m^2

20)

Explian the question : Known wavelength $\lambda_1 = 590$ nm
- Unknown wavelength $\lambda_2 = ?$
- Central maxima coincide (so zero order overlaps).
- The 3rd bright fringe of known light coincides with the 4th bright fringe of unknown light.
- We need to find λ_2

Concept : For YDSE, position of nth bright fringe is :

$$x_n = n\frac{\lambda D}{d}$$

(where D = distance to screen, d = slit separation)

If fringes coincide at the same position :

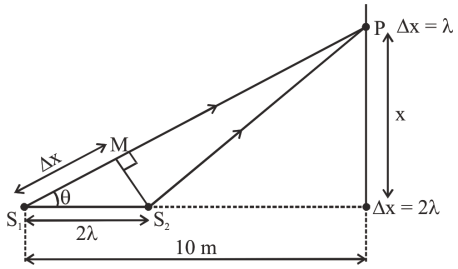
$n_1\lambda_1 = n_2\lambda_2$

$$\frac{n_1 D\lambda_1}{d} = \frac{n_2 D\lambda_2}{d}$$

$$\Rightarrow n_1\lambda_1 = n_2\lambda_2 \Rightarrow \lambda_2 = \frac{n_1}{n_2} \times \lambda_1$$

$$= \frac{3}{4} \times 590\,nm = 442.5 \text{ nm}$$

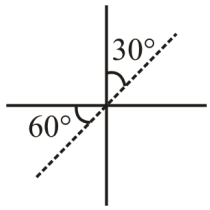
21)



$$\cos\theta = \frac{\Delta x}{2\lambda} = \frac{\lambda}{2\lambda} = \frac{1}{2} \Rightarrow \theta = 60°$$

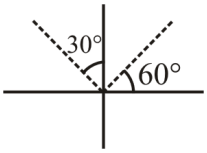
$$\Rightarrow \frac{x}{10} = \tan 60° \Rightarrow x = 10\sqrt{3}\,m$$

22)



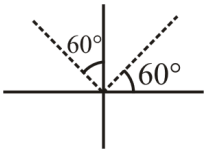
(i) θ = 0°

$$I_1 = \frac{I_0}{2}\cos^2 0° = \frac{I_0}{2}$$



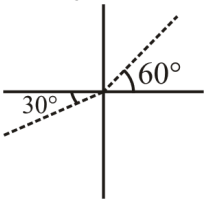
(ii) θ = 60°

$$I_2 = \frac{I_0}{2}\cos^2(60°) = \frac{I_0}{8}$$



(iii) θ = 90°

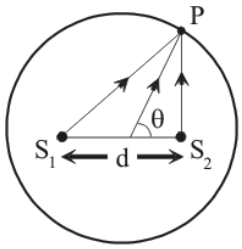
$$I_3 = \frac{I_0}{2}\cos^2 90° = 0$$



(iv) $\theta = 30°$

$$I_4 = \frac{I_0}{2}\cos^2 30° = \frac{3I_0}{8}$$

$I_1 > I_4 > I_2 > I_3$



23) Here path difference at a point on the circle is given by $\Delta x = d\cos\theta$ ..... (i)

For maxima at

$\Delta x = n\lambda$ ..... (ii)

From equation (i) and (ii)

$$n\lambda = d\cos\theta \Rightarrow \theta\cos^{-1}\left(\frac{n\lambda}{d}\right) = \cos^{-1}\left(\frac{4\lambda}{d}\right)$$

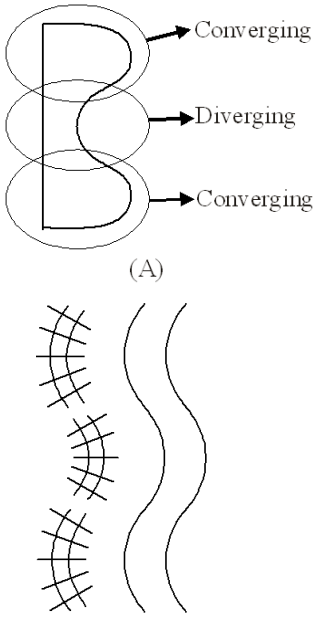
24) **Question Explanation :**

To find how wavefront of light changes upon striking the boundary of medium.

Concept :

This question is based on wavefront of light, Huygens principle.

Solution :



Hence,

Option (1) is correct

25)

$\lambda = \frac{h}{p} \Rightarrow \lambda \alpha \frac{1}{p}$

As momentum in creases, the wavelength decreases.

26)

Reaction:

$^{239}_{94}Pu \rightarrow ^{235}_{92}U + He$

Let,

Mass of α-particle $= m_\alpha = 4$

Mass of daughter nucleus $= m_d = 235$

Energy carried by α-particle:

$E_\alpha = Q.\frac{m_d}{m_d + m_\alpha}$

$E_\alpha = Q.\frac{235}{235+4} = Q.\frac{235}{239}$

$\frac{235}{239} \times 100 = 98.32\%$

27)

Energy in Joule

$= 200 \times 10^6 \times 1.6 \times 10^{-19}$

$= 3.2 \times 10^{-11}$ Joule

$P = \frac{Energy}{time}$

Energy Released will be $= n \times 3.2 \times 10^{-11}$

$2 \times 10^3 = n \times 3.2 \times 10^{-11}$

$n = 6.25 \times 10^{13}$

28) Required reaction is $Cu^{64} \rightarrow Cu^{63} + n^1$

Q value = (63 × 8.75 – 64 × 8.73)

= 551.25 – 558.72 = –7.47 MeV

29)

Energy required $= (\Delta m)C^2$

$= [M_P + M_n - M_d]C^2$

= [1.0078 + 1.0087 – 2.0141] × 931 MeV

= 2.2344 MeV

$= 2.2344 \times 10^6 \times 1.6 \times 10^{-19}$ J

but $E = h\nu \Rightarrow \nu = \frac{E}{h}$

So, $\nu = \frac{2.2344 \times 1.6 \times 10^{-13}}{6.6 \times 10^{-34}}$

$= 5.4 \times 10^{20}$ Hz

30)

$n_1 h\nu = I_1$
$n_2 h(3\nu) = 4I_1$
$\frac{n_1}{n_2} = \frac{3}{4}$
$\frac{36}{i} = \frac{3}{4}$
i = 48 mA

31) Here, P = 100 W, λ_1 = 1 nm, λ_2 = 500 nm Let n_1 and n_2 be the number of photons of X-rays and visible light emitted from the two sources
$n_1 \frac{hc}{\lambda_1} = n_2 \frac{hc}{\lambda_2}$ or $\frac{n_1}{\lambda_1} = \frac{n_2}{\lambda_2}$ or $\frac{n_1}{n_2} = \frac{\lambda_1}{\lambda_2} = \frac{1}{500}$

32)

Threshold wavelength $\lambda_0 = 5 \times 10^{-7}$ m.
Initial wavelength $\lambda_1 = 3 \times 10^{-7}$ m
Since $\lambda_1 < \lambda_0$, photoemission occurs and stopping potential is V_0.
When wavelength is doubled, $\lambda_2 = 6 \times 10^{-7}$ m
Now $\lambda_2 > \lambda_0$, so photon energy is less than work function and no photoelectric emission occurs.
Hence stopping potential becomes zero.

33)

248 nm = 1240/248 eV = 5 eV
310 nm = 1240/310 eV = 4 eV
$\frac{K.E_1}{K.E_2} = \frac{4}{1} = \frac{5eV - W}{4eV - W}$
$\Rightarrow$ 16 – 4W = 5 – W $\Rightarrow$ 11 = 3W
$\Rightarrow$ $W = \frac{11}{3} = 3.67 \text{ eV} \cong 3.7 \text{ eV}$

34)

$\omega_1 = 5 \times 10^{15}$ & $\omega_2 = 8 \times 10^{15}$
for k_{max} for ω should be maximum
$\omega = 8 \times 10^{15} = 2\pi f$
$\Rightarrow f = \frac{8 \times 10^{15}}{2\pi}$
$E = hf = \frac{6.6 \times 10^{-34} \times 8 \times 10^{15}}{2\pi \times 1.6 \times 10^{-19}} eV$
$E \simeq 5.25 eV$
$k_{max} = E - \phi$
= (5.25 – 2) eV = 3.25 eV

35) Energy produced by reactor in 1 hr
$= 500 \times 10^6 \times 3600$ J
$= 1.8 \times 10^{12}$ J

nuclear energy released = 3.6×10^{12} J

mass lost = $\frac{\text{Energy}}{c^2} = \frac{3.6 \times 10^{12}}{(3 \times 10^8)^2}$

$= 4 \times 10^{-5}$ kg/hr. = 0.04 gm/hr.

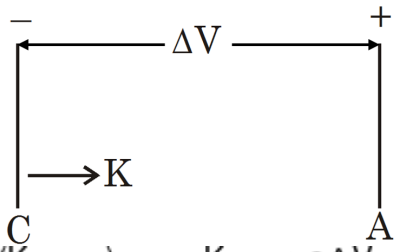
36) $Ne^{20} \rightarrow 2He^4 + C^{12}$

$8.03 \times 20 \rightarrow 2 \times 7.07 \times 4 + 7.68 \times 12$

Δ value = $(BE)_{product} - (BE)_{react}$ = –11.88 MeV

So 11.88 MeV will have to be supplied.

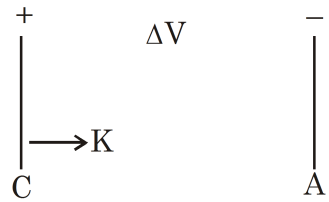
37) $(KE)_{max} = \frac{hc}{\lambda} - \phi \Rightarrow \frac{1240}{200} - 4.7 = 1.5eV$



$(K_{Anode})_{max} = K_{max} + e\Delta V$

$= 1.5 + 1 = 2.5$ eV

$(K_{Anode})_{min} = K_{min} + e\Delta V = 1$ eV



$(K_{Anode})_{max} = 1.5 - 0.5 = 1eV$

$(K_{Anode})_{min} = 0.5 - 0.5 = 0$

38) $\lambda = \frac{h}{\sqrt{2mE}}$

$\lambda^2 = \frac{h}{2mE} \Rightarrow \lambda^2 = \frac{h}{2m} E^{-1}$

$\lambda = \left(\frac{h}{2m}\right)^{\frac{1}{2}} E^{\frac{-1}{2}}$

$\log \lambda = \log \left(\frac{h}{2m}\right)^{\frac{1}{2}} + \log (E)^{\frac{-1}{2}}$

$\log \lambda = -\frac{1}{2} \log E + \frac{1}{2} \log \left(\frac{h}{2m}\right)$

y = – mx + C

39) In one fusion reaction energy released is 26 MeV.

In one fission reaction energy released is 200 MeV.

Energy per nucleon in fission (0.85 MeV) << Energy per nucleon in fussion (6.5 MeV) the abundance of hydrogen make it unlimited source of energy.

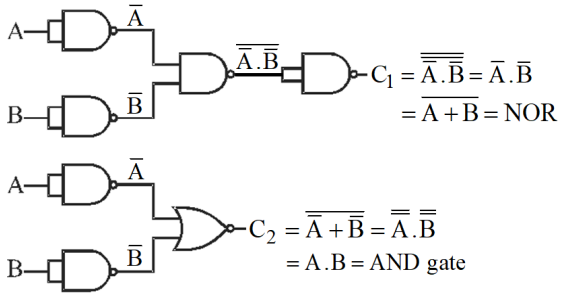
40)

For Ge crystal, $\Delta Eg = 0.7$ eV

$\lambda \leqslant \frac{hc}{\Delta Eg}$ or $\lambda_{max} = \frac{hc}{\Delta Eg}$

$= \frac{12400}{0.7} = 1771.42\,nm$

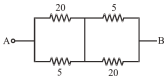
41)



42)

Option (1)

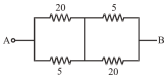
$V_A > V_B$, $S_1 \Rightarrow$ open, $S_2 \Rightarrow$ Closed



$\Rightarrow R = 8\Omega$

Option : (2)

$V_A > V_B$, $S_1 \Rightarrow$ closed, $S_2 \Rightarrow$ Open



$\Rightarrow R = 8\Omega$

Option : (3)

Same as option (2), $R = 8\Omega$

correct choice = All of above

43)

Peak to peak voltage = $2\,V_0 = 310$ V

So, $V_0 = 155$ V

V_{rms} (in HWR) $= \frac{V_0}{2} = \frac{155}{2} = 77.5$ volt

44)

According to zener diode theory

45)

Explain question:

The question tests the understanding of the assertion and reason related to the nature of NAND and NOR gates.

Concept:
NAND and NOR gates
Solution:
The assertion is true because NAND and NOR gates are indeed called digital building blocks.
The reason is also true, as NAND and NOR gates can be used to create all basic gates.
Final answer : 1